SyteScan: A Multimodal Intelligent System for Interior Site Analysis and Contractor Discovery

A

Synopsis Report

Submitted in Partial Fulfillment of the Requirements for the Minor Degree/Specialization in

Multimodal Intelligent Systems

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1 Introduction

Interior project planning is often inefficient due to incomplete communication between customers and interior contractors. Clients typically struggle to clearly describe their requirements, budget constraints, and site conditions, while contractors receive insufficient information to prepare accurate quotations. This results in delays, miscommunication, and poor decision-making during contractor selection [1].

With the increasing adoption of digital platforms and artificial intelligence, there is a significant opportunity to improve this process through intelligent systems that can analyze multiple forms of input data—images, textual descriptions, and structured information [2].

The proposed system, **SyteScan**, aims to provide an intelligent platform that collects interior site information from users, analyzes it using multimodal processing techniques, and generates a structured interior project report. This report can subsequently be used for contractor discovery and quotation, streamlining the entire interior project lifecycle.

2 Problem Statement

Customers planning interior work face difficulty in documenting site conditions, communicating design requirements, and identifying suitable contractors. Most existing platforms only provide marketplace listings without assisting users in preparing a structured project description.

Without accurate project documentation, contractors must rely on incomplete information, leading to inaccurate quotes and inefficient decision-making. Therefore, there is a pressing need for a system that can:

- Collect comprehensive site information through multiple modalities;
- Analyze images, textual descriptions, and structured data intelligently;
- Generate a structured project report that simplifies interior planning and contractor discovery.

3 Objectives

The primary objectives of the SyteScan project are:

1. To design a system that collects interior site information including photographs, measurements, requirements, and budget.
2. To analyze multiple data modalities—images and textual descriptions—using intelligent processing techniques such as computer vision and natural language processing [3, 4].

3. To automatically generate a structured interior site report summarizing project requirements and recommendations.
4. To provide a conceptual foundation for an interior marketplace where contractors can review project reports and submit quotations.
5. To demonstrate the practical application of multimodal intelligent systems in real-world interior planning [5].

4 Proposed System

The SyteScan system comprises two conceptual modules:

4.1 Module 1: Site Intelligence & Report Generation

This core module collects user inputs—site photographs, room measurements, design requirements, and budget constraints—and processes them through a multimodal analysis pipeline. The output is a structured **Interior Site Report** containing key project details, design insights drawn from image analysis, and actionable recommendations.

4.2 Module 2: Interior Marketplace (Conceptual)

The generated site report can be published to an interior marketplace where registered interior companies can view project requirements, analyze the site report, and submit quotations. Users can then compare contractor quotes, portfolios, ratings, and reviews before selecting a contractor.

- **Signal:** Contractor Discovery Confidence
- **Impact:** Quoted scope quality drops when room constraints are inferred from sparse imagery
- **Action:** Require geometry validation before quote release

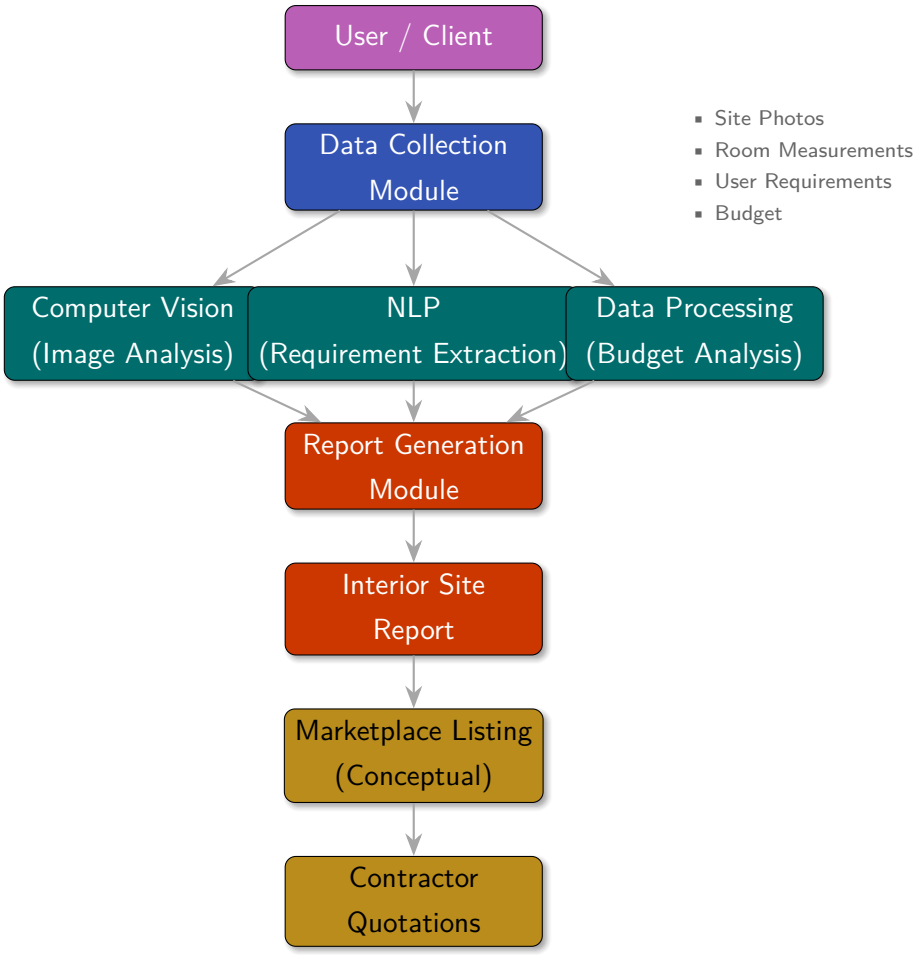

Figure 1: SyteScan — High-Level System Architecture

5 Scope of the Mini Project

Given the academic scope of the Minor Specialization Mini-Project, the **implementation primarily focuses on the Site Intelligence and Report Generation module**. The marketplace component is presented at the conceptual design level.

The implemented system includes:

- A web-based user interface for site data collection;
- Processing of site photographs using computer vision techniques;
- Extraction of requirements from textual descriptions via NLP;
- Automated generation of a structured interior project report.

6 Methodology

The system follows a **multimodal processing pipeline** where different types of input data are routed through specialized analysis modules before being synthesized into a unified report. Figure 2 illustrates the end-to-end methodology.

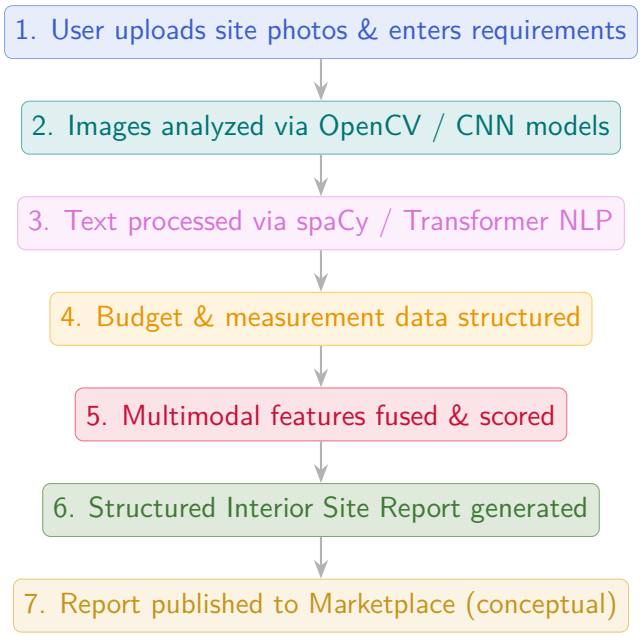

Figure 2: End-to-End Methodology Pipeline

7 Technologies Used

Table 1 summarizes the technology stack selected for each component of the system.

Table 1: Technology Stack

Component	Technology
Frontend	Next.js (React framework)
Backend API	FastAPI (Python)
Database	PostgreSQL
Image Processing	OpenCV, Python (CNN-based feature extraction)
NLP Processing	spaCy, Transformer models (BERT / DistilBERT)
Storage	Object storage for site images (e.g., AWS S3 / MinIO)

8 Expected Outcomes

1. A working prototype capable of collecting interior site information through a web interface.
2. Automated analysis of site photographs using computer vision to extract room features, lighting conditions, and surface materials.

3. NLP-driven extraction and structuring of user requirements from free-text descriptions.
4. Generation of a comprehensive Interior Site Report summarizing project scope, visual analysis, and budget insights.
5. A conceptual interior marketplace model that demonstrates contractor discovery and quotation workflows.

9 Relevance to Minor Specialization

SyteScan directly aligns with the **Multimodal Intelligent Systems** minor specialization by integrating techniques from multiple domains studied during the program. Table 2 maps each minor course to its application within the project.

Table 2: Mapping of Minor Courses to Project Application

Minor Course	**Application in SyteScan**
Information Retrieval	Contractor discovery, project report indexing, and quote ranking
Natural Language Processing	Extraction and structuring of interior requirements from user descriptions
Computer Vision	Analysis of site photographs—room layout, materials, lighting, and condition
Social Network Analysis	Contractor reputation modeling, review networks, and interaction analysis

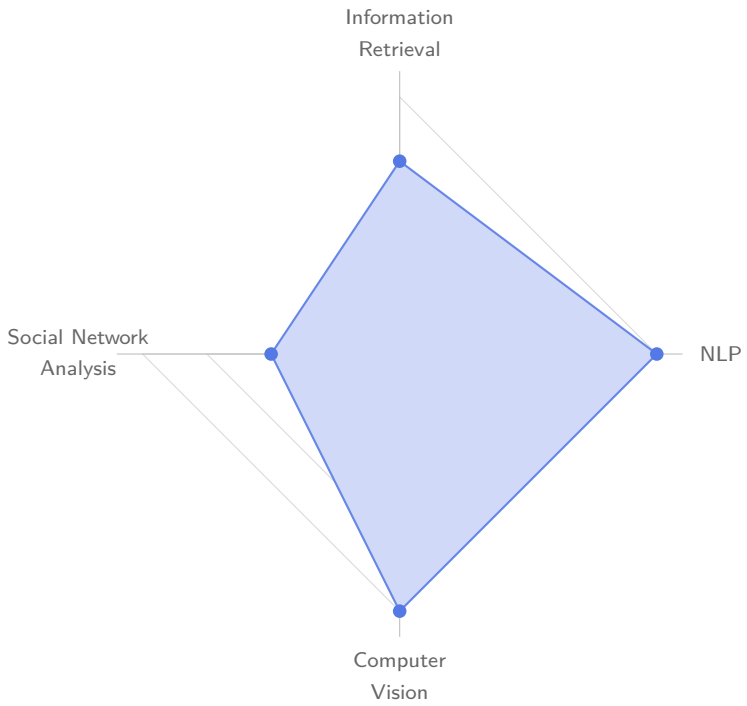

Figure 3: Relevance Intensity of Minor Specialization Courses to SyteScan

10 Conclusion

The proposed system, SyteScan, demonstrates the potential of multimodal intelligent systems in improving interior project planning and contractor discovery. By integrating computer vision for site image analysis, natural language processing for requirement extraction, and structured data analysis for budget and measurement handling, the system aims to generate actionable project insights that simplify communication between customers and interior professionals.

The project provides both academic relevance—through its alignment with the Multimodal Intelligent Systems minor specialization—and practical applicability in the growing digital interior services market.

References

[1] T. Baltrušaitis, C. Ahuja, and L.-P. Morency, "Multimodal machine learning: A survey and taxonomy," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 2, pp. 423–443, 2019.

[2] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

[3] D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed. Pearson, 2021.

[4] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004.

[5] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge University Press, 2008.